

Ameer Hassan

ameer.hassan726@gmail.com | +1 (973) 981-1779 | Woodland Park, NJ 07424 | [LinkedIn](#) | [GitHub](#)

EDUCATION

New Jersey Institute of Technology - Newark, New Jersey

Expected Graduation Date: May 2028

Bachelor of Science, Computer Science, Minor in Applied Mathematics

Cumulative GPA: 3.94

- Relevant Coursework: Java Data Structure & Algorithms, Python Essentials, Introduction to Java, Internet Applications, Calculus I & II, Quantitative Analysis Application Business, Macroeconomics
- Albert Dorman Honors Scholar & Dean's List: All Semesters

TECHNICAL SKILLS

Programming Languages: Python | JavaScript | TypeScript | Java | C++ | HTML/CSS | SQL

Frameworks & Libraries: React.js | Node.js | TensorFlow | Flask | Tailwind CSS | OpenAI API | WebRTC | Convex

Developer Tools: Git | GitHub | VS Code | Jupyter Notebook | Google Colab | Anaconda | Postman

Technologies & Platforms: WebSockets | Judge0 IDE | Gemini 1.5 Flash | MySQL

EXPERIENCE

Software Engineering Resident, Headstarter, Remote

Jul 2025 - Present

- Building over 14 machine learning, AI-engineering, and full-stack projects in fast-paced software team environments.
- Developing 5+ **neural networks** in Python and 11 applications in TypeScript on AWS and Vercel, managing both development and production environments.
- Implementing **LLM chaining, hyperparameter tuning**, and fine-tuning across 10+ large language models, optimizing for latency and accuracy.
- Being mentored by engineers from Google (Machine Learning, Kubernetes), Two Sigma, Tesla, Figma, and Citadel.

AI/ML Research Assistant, NJIT YWCC, Newark, NJ

Jun 2025 - Dec 2025

- Selected as 1 of 3 undergraduates to join Professor James Wang's machine learning research team, contributing to a project aimed at academic publication. Tasks include reviewing over **25+ peer-reviewed papers** and preprocessing datasets with **10,000+ entries**.
- Collaborating with a 5-member team to design and test predictive models using Python, scikit-learn, and TensorFlow.

Technical Peer Mentor, NJIT Learning Committee, Newark, NJ

Jun 2025 - May 2026

- Coached **15+ first-year CS students** in **Java, Python**, and debugging, providing one-on-one coding sessions and improving project completion rates by **30%**.
- Organized **3 campus-wide academic and technical events**, including hackathon prep workshops, boosting student participation by **40%** compared to the previous semester.

PROJECTS

[Discord – Real-Time Chat & Video](#) | React | Convex | WebSockets | WebRTC | Tailwind CSS

- Developed a full-stack Discord clone enabling real-time chat and video communication for **100+ concurrent users**
- Engineered a **responsive UI** with reusable React components, reducing frontend development time by **30%** and improving maintainability.
- Integrated **WebRTC** to support low-latency video/audio calls (**<200ms**) and built real-time backend features including live chat, user presence, and dynamic channel creation.

[AI Code Editor](#) | TypeScript | React | Node.js | OpenAI API

- Built AI-powered features on top of the **Judge0 IDE**, including a live chat assistant that provides code explanations, bug fixes, and real-time help for compile-time errors.
- Integrated **inline code chat** and **code-aware autocomplete**, enabling users to select code segments and receive contextual AI responses with **<300ms latency**.
- Developed a bug detection tool using AI to identify common logic and syntax errors, and implemented **PostgreSQL** to log user sessions and store code history, improving debugging efficiency and personalization.

[Brain Tumor Classification System – Deep Learning Pipeline](#) | Python | TensorFlow | Flask

- Built a brain tumor classification system using **convolutional neural networks (CNNs)** and transfer learning, achieving over **90% accuracy** on MRI scan datasets.
- Designed and trained custom deep learning architectures with TensorFlow and Keras, optimizing performance through data augmentation and regularization techniques.
- Integrated **Gemini 1.5 Flash** to generate natural language explanations for model predictions, improving model interpretability for non-technical users.

LEADERSHIP EXPERIENCE

Marketing Director - HD Bags LLC

May 2023 - Present

Boys Youth Committee Board Member - Islamic Center of Passaic County

Apr 2025 - Present